

Wenzhi Zhao

PhD Student in Earth and Environmental Science

Asian School of the Environment | Earth Observatory of Singapore
Nanyang Technological University, Singapore

wenzhi001@e.ntu.edu.sg | [Google Scholar](#) | [ORCID](#) | [GitHub](#)

Research Interests

Earthquake physics and rock mechanics; fault friction and rate-and-state friction laws; laboratory shear experiments; earthquake cycle simulations; coupling between fault slip and permeability.

Education

PhD in Earth and Environmental Science | Aug 2025 - Present

Nanyang Technological University, Singapore
Advisor: Luca Dal Zilio

MEng in Geotechnical Engineering | Sep 2021 - Jun 2024

Tongji University, China
Supervisor: Fengshou Zhang

BEng in City Underground Space Engineering | Sep 2017 - Jun 2021

Southwest Jiaotong University, China

Publications

Wenzhi Zhao, Mengke An, Luca Dal Zilio, Luanxiao Zhao, Hehua Zhu, Xianda Shen, Quan Gan, Fengshou Zhang, and Derek Elsworth (2025). Coupled evolution of seismicity and permeability in fault zones. *Communications Earth & Environment*, 6, 918. [DOI](#)

Fengshou Zhang, **Wenzhi Zhao**, Mengke An, Xianda Shen, Jizhou Tang, Luanxiao Zhao, Hai Liu, Derek Elsworth, Hehua Zhu, and Manchao He (2024). Shallow Moonquake Mechanisms Illuminated by Rheologic Characteristics of Basaltic Gouges. *Journal of Geophysical Research: Planets*, 129(11), e2024JE008370. [DOI](#)
[Associated dataset: Dryad](#)

Hu Luo, Jianjun Tong, **Wenzhi Zhao**, Guo Li, Jinsheng Liu, and Yanshan Cai (2022). Degradation test of mechanical properties of sprayed concrete by sulfate erosion under high and variable temperature conditions. *Sichuan Construction*, 42(1), 193-195. (In Chinese.)

Research Experience

Earthquake physics and rock mechanics

Doctoral research, Nanyang Technological University | Aug 2025 - Present

- Research interests include friction laws, laboratory shear experiments, and earthquake cycle simulations under the supervision of Luca Dal Zilio.

Coupled evolution of seismicity and fault permeability

- Synthesized experimental, theoretical, and numerical studies of friction and permeability evolution in faults and fractures, with implications for reservoirs and caprock integrity.
- Examined structural controls and coupling across coseismic and interseismic processes. First-author review published in *Communications Earth & Environment* in 2025.

Shallow moonquake mechanisms in basaltic gouges

Research project | Jun 2022 - Jun 2024

- Prepared synthetic fault gouges informed by lunar crustal mineralogy and Chang'e-5 samples; investigated how temperature, stress, humidity, and microstructure affect frictional behavior.
- Interpreted experimental results alongside lunar temperature profiles to examine conditions for brittle faulting and shallow moonquakes. Published in *Journal of Geophysical Research: Planets* in 2024.

Sprayed concrete degradation in thermal and sulfate environments

Research project | Jun 2020 - Jun 2021

- Investigated concrete degradation under high and variable temperatures and sulfate exposure in relation to tunnel stability; contributed to a Chinese-language publication in 2022.

Honors and Awards

Tongji University Academic Scholarship | September 2021, 2022, and 2023

Provincial First Prize, China Undergraduate Mathematical Contest in Modeling | November 2019

Technical Skills

Computing: MATLAB and Python for data analysis and visualization; discrete element modeling with PFC and MatDEM; experience applying machine learning to partial differential equations.

Laboratory methods: Rock sample preparation; biaxial direct shear; high-temperature and high-pressure triaxial testing; ring shear (ICL-2); high-speed shear and Hopkinson compression bar experiments.